Supplementary Material

This file contains the following appendices:

1. **Appendix A** contains material relevant to RQ-1. Specifically:
   - Table A.1 presents the types of anomalies in thirteen publicly available datasets.
   - Table A.2 presents the percentage of normal data, and the three anomaly categories in each dataset.
   - Table A.3 presents details of each anomaly category.
   - Table A.4 presents details of ADS for each anomaly category.
   - Table A.5 presents details of the SS for each dataset.
   - Table A.6 presents details of the SS for each RL algorithm.
   - Table A.7 presents details of the SS for each Gen AI algorithm.
2. **Appendix B** contains material relevant to RQ-2. Specifically:
   - Table B.1 provides a summary of the value-based RL approaches used in cybersecurity.
   - Table B.2 provides a summary of the policy gradient-based RL approaches used in cybersecurity.
   - Table B.3 provides a summary of the hybrid actor-critic-based RL approaches used in cybersecurity.
   - Table B.4 provides a summary of the state-of-the-art studies based on RL algorithms for cybersecurity.
3. **Appendix C** contains material relevant to RQ-3. Specifically:
   - Table C.1 provides a summary of GAN-based generative AI approaches used in cybersecurity.
   - Table C.2 presents VAE-based generative AI approaches used in cybersecurity.
   - Table C.3 summarizes transformer and large language model (LLM)-based generative AI techniques used in cybersecurity.
   - Table C.4 outlines hybrid and emerging generative AI approaches for cybersecurity.
   - Table C.5 provides a summary of the state-of-the-art studies based on Gen AI algorithms for cybersecurity.

Appendix A. Tables Relevant to RQ-1

Table A.1: Publicly available datasets for anomaly detection.

Dataset Dataset	No. of categories	Types of anomalies
UNSW–NB15	9	Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms
SWaT	2	Normal and abnormal
MIMIC–III	-	Policy violation
CIC–IDS2017	15	DoS, PortScan, Web (XSS, SQLi), Heartbleed, etc.
N–BaIoT	2	Mirai, BASHLITE
Bot–IoT	4	DDoS, DoS, Reconnaissance, Information Theft
ECU–IoHT	4	ARP spoofing, DoS, Smurf, PortScan
ToN–IoT	9	DoS, DDoS, Backdoor, Injection, Password, Ransomware, XSS, MITM
IoT–23	12	Okiru, Mirai, Torii, C&C–HeartBeat
CIC IoT 2023	7	DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai
CIC IoT DIAD 2024	7	DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai
CIC IoMT 2024	5	DDoS, DoS, Recon, MQTT, Spoofing
DataSense: CIC IIoT 2025	7	Reconnaissance, DoS, DDoS, Web, MITM, Brute Force, Malware (Mirai) + Benign; 49 attack types

Table A.2: Distribution of anomalies in datasets.

Dataset	Normal / Benign	First largest anomaly type	Second largest anomaly type	Third largest anomaly type
UNSW–NB15	87%	Generic (8%)	Exploits (2%)	DoS (1%) Reconnaissance (1%) Fuzzers (1%)
CICIDS–2017	62%	Port Scan (12%)	DoS–Hulk (11%)	DDoS–LOIC (7%)
BoT–IoT	0%	DDoS (53%)	DoS (45%)	Reconnaissance/ Service Scan (2%)
ToN–IoT	4%	Scanning (32%)	DDoS (28%)	DoS (15%)
IoT–23	11%	Port Scan (56%)	Okiru (22%)	DDoS (11%)
CIC–IoT–DIAD	25%	Web–based (32%)	Brute Force (28%)	Mirai (3%) DoS (3%) DDoS (3%) Spoofing (3%) Reconnaissance (3%)
DataSense: CIC IIoT Dataset 2025	$< 0\%$	DDoS (63%)	DoS (29%)	Man-in-the-Middle (7%)

Table A.3: Details of anomaly categories.

Anomaly Category	Tactics	Techniques	Description
Adversarial threats	AML.T0008, AML.T0011	AML.T0002, AML.T0016	AI-generated malicious content, automated attacks, adversarial threats, evasion attacks, data poisoning, malware synthesis, polymorphic malware, adversarial evasion, zero-day attacks
AI-generated threats	AML.T0017, AML.T0013	AML.T0003, T1204	Automated phishing, prompt injection, code exploits, deepfakes, misinformation, LLM-aided propaganda, jailbreaking LLMs, AI-generated malware
Behavioural anomalies	TA0001, TA0002, TA0005, TA0007, TA0009	T1078, T1082, T1003, T1036, T1055	APTs, insider threats, evasive behaviour, lateral movement, MITM, ransomware, UEBA scenarios
Data leakage anomalies	TA0010, TA0009	T1005, T1041	Privilege misuse, data exfiltration
Host-based anomalies	TA0002, TA0003, TA0005	T1059, T1547, T1055, T1036	Intrusion attempts, privilege abuse, host-based detection
Intrusion anomalies	TA0001, TA0002, TA0004, TA0005, TA0008, TA0010	T1059, T1547, T1210, T1027, T1036	Brute force, shellcode, DoS, GPS spoofing, AIS manipulation, botnets, privilege escalation, zero-day exploits

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Table A.3: Details of anomaly categories.

Anomaly Category	Tactics	Techniques	Description
Network-based anomalies	TA0011, TA0010, TA0007	T1041, T1071.001, T1040, T1046	DoS, U2R, R2L, spoofing, IoT attacks, botnet traffic, anomalous flows, adversarial traffic
Policy violation anomalies	TA0002, TA0005	T1204, T1036	Data tampering, integrity violations
Synthetic traffic anomalies	TA0001, TA0011	T1090, T1071.001	Synthetic flows, anomalous injections
Unknown/ Adaptive threats	TA0001, TA0004, AML.T0014	T1078, T1211, AML.T0011	APTs, IoT botnets, protocol anomalies, false data injection, reconnaissance, ransomware, MITM, password brute force, zero-day

End of table

Table A.4: ADS Details.

Anomaly Category	PR	RE	DF	ADS	Description	
Adversarial threats	9	9	8	9	8.75	Adversarial ML attacks exploit vulnerabilities in AI models; subtle perturbations hinder detection; prevention requires robust training; removal needs retraining; defence requires explainability/testing.
AI-generated threats	8	9	8	9	8.5	Includes deepfakes and synthetic phishing; hard to detect in real-time; prevention via model alignment; removal hard due to viral spread; defence needs GenAI detection.
Behavioural anomalies	8	6	5	7	6.5	Hard to detect due to blending with legitimate actions; moderate prevention with access controls; removal easier (no malware); defence needs UEBA.
Data leakage anomalies	8	7	7	8	7.5	Hard when encrypted/covert; prevention needs DLP; removal impossible post-exfiltration; defence needs encryption/monitoring.
Host-based anomalies	7	6	7	6	6.5	Mimics legitimate apps; prevention via whitelisting/EDR; removal may require re-imaging; defence via host IDS/EPP.
Intrusion anomalies	7	6	6	7	6.5	Multi-stage attacks; detection via NIDS/HIDS; prevention via segmentation; removal depth-dependent; defence layered.
Network-based anomalies	6	5	4	6	5.25	Detected via stats/flow; prevention via segmentation; removal often irrelevant; defence requires robust monitoring.
Policy violation anomalies	4	4	3	4	3.75	Easier to detect via logs; prevention with rules; removal straightforward; defence = enforcement/monitoring.
Synthetic traffic anomalies	6	5	3	6	5	Mimics real traffic; detection moderate; prevention/removal easier; defence = traffic baselines.
Unknown/ Adaptive threats	9	9	9	9	9	Evolving threats bypass signatures; very hard detection; prevention without AI difficult; removal may need rebuild; defence = zero-trust.

Table A.5: SS for datasets.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
Adversarial threats	BoT-IoT	0	0	0	0	0
AI-generated threats	BoT-IoT	0	0	0	0	0
Behavioral	BoT-IoT	0	0	0	0	0
Data leakage	BoT-IoT	0	0	0	0	0
Host-based	BoT-IoT	3	3	1	3	2.5
Intrusion anomalies	BoT-IoT	3	3	3	3	3
Network-based	BoT-IoT	3	3	3	3	3
Policy violation	BoT-IoT	0	0	0	0	0
Synthetic traffic	BoT-IoT	0	0	0	0	0
Unknown or Adaptive threats	BoT-IoT	0	0	0	0	0
Adversarial threats	CICIDS2017	3	3	2	2	2.5
AI-generated threats	CICIDS2017	0	0	0	0	0
Behavioral	CICIDS2017	3	3	3	3	3
Data leakage	CICIDS2017	0	0	0	0	0
Host-based	CICIDS2017	0	0	0	0	0
Intrusion anomalies	CICIDS2017	3	3	3	3	3
Network-based	CICIDS2017	0	0	0	0	0

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Table A.5: SS for datasets.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
Policy violation	CICIDS2017	0	0	0	0	0
Synthetic traffic	CICIDS2017	0	0	0	0	0
Unknown or Adaptive threats	CICIDS2017	3	3	2	3	2.75
Adversarial threats	CICIDS2018	3	3	1	3	2.5
AI-generated threats	CICIDS2018	0	0	0	0	0
Behavioral	CICIDS2018	0	0	0	0	0
Data leakage	CICIDS2018	0	0	0	0	0
Host-based	CICIDS2018	0	0	0	0	0
Intrusion anomalies	CICIDS2018	0	0	0	0	0
Network-based	CICIDS2018	3	3	3	3	3
Policy violation	CICIDS2018	0	0	0	0	0
Synthetic traffic	CICIDS2018	0	0	0	0	0
Unknown or Adaptive threats	CICIDS2018	3	3	3	2	2.75
Adversarial threats	KDDCup99	0	0	0	0	0
AI-generated threats	KDDCup99	0	0	0	0	0
Behavioral	KDDCup99	0	0	0	0	0
Data leakage	KDDCup99	0	0	0	0	0
Host-based	KDDCup99	0	0	0	0	0
Intrusion anomalies	KDDCup99	0	0	0	0	0
Network-based	KDDCup99	3	2	1	3	2.25
Policy violation	KDDCup99	0	0	0	0	0
Synthetic traffic	KDDCup99	0	0	0	0	0
Unknown or Adaptive threats	KDDCup99	3	2	1	2	2
Adversarial threats	NSL-KDD	3	3	3	2	2.75
AI-generated threats	NSL-KDD	0	0	0	0	0
Behavioral	NSL-KDD	3	3	3	3	3
Data leakage	NSL-KDD	0	0	0	0	0
Host-based	NSL-KDD	0	0	0	0	0
Intrusion anomalies	NSL-KDD	3	3	3	3	3
Network-based	NSL-KDD	3	3	3	3	3
Policy violation	NSL-KDD	0	0	0	0	0
Synthetic traffic	NSL-KDD	0	0	0	0	0
Unknown or Adaptive threats	NSL-KDD	3	3	1	2	2.25
Adversarial threats	ydata-synthetic	3	3	3	2	2.75
AI-generated threats	ydata-synthetic	0	0	0	0	0
Behavioral	ydata-synthetic	3	3	3	2	2.75
Data leakage	ydata-synthetic	0	0	0	0	0
Host-based	ydata-synthetic	0	0	0	0	0
Intrusion anomalies	ydata-synthetic	3	3	3	2	2.75
Network-based	ydata-synthetic	0	0	0	0	0
Policy violation	ydata-synthetic	3	3	1	3	2.5
Synthetic traffic	ydata-synthetic	0	0	0	0	0
Unknown or Adaptive threats	ydata-synthetic	3	3	1	1	2
Adversarial threats	TON-IoT	0	0	0	0	0
AI-generated threats	TON-IoT	0	0	0	0	0
Behavioral	TON-IoT	0	0	0	0	0
Data leakage	TON-IoT	0	0	0	0	0
Host-based	TON-IoT	0	0	0	0	0
Intrusion anomalies	TON-IoT	3	3	3	3	3
Network-based	TON-IoT	3	3	3	3	3
Policy violation	TON-IoT	0	0	0	0	0
Synthetic traffic	TON-IoT	0	0	0	0	0
Unknown or Adaptive threats	TON-IoT	3	3	1	3	2.5
Adversarial threats	UNSW-NB15	3	3	2	2	2.5
AI-generated threats	UNSW-NB15	0	0	0	0	0
Behavioral	UNSW-NB15	3	3	2	2	2.5
Data leakage	UNSW-NB15	0	0	0	0	0
Host-based	UNSW-NB15	0	0	0	0	0
Intrusion anomalies	UNSW-NB15	3	3	3	3	3
Network-based	UNSW-NB15	3	3	3	3	3
Policy violation	UNSW-NB15	0	0	0	0	0
Synthetic traffic	UNSW-NB15	0	0	0	0	0
Unknown or Adaptive threats	UNSW-NB15	3	3	2	3	2.75

End of table

Table A.6: SS for RL algorithms.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
Adversarial threats	A3C	0	0	0	0	0
AI-generated threats	A3C	0	0	0	0	0
Behavioral	A3C	0	0	0	0	0
Data leakage	A3C	0	0	0	0	0
Host-based	A3C	0	0	0	0	0
Intrusion anomalies	A3C	0	0	0	0	0
Network-based	A3C	0	0	0	0	0
Policy violation	A3C	0	0	0	0	0
Synthetic traffic	A3C	0	0	0	0	0
Unknown or Adaptive threats	A3C	3	1	2	3	2.25
Adversarial threats	AC	0	0	0	0	0
AI-generated threats	AC	0	0	0	0	0
Behavioral	AC	0	0	0	0	0
Data leakage	AC	0	0	0	0	0
Host-based	AC	0	0	0	0	0
Intrusion anomalies	AC	3	1	1	2	1.75
Network-based	AC	0	0	0	0	0
Policy violation	AC	0	0	0	0	0
Synthetic traffic	AC	0	0	0	0	0
Unknown or Adaptive threats	AC	0	0	0	0	0
Adversarial threats	DDPG	0	0	0	0	0
AI-generated threats	DDPG	0	0	0	0	0
Behavioral	DDPG	3	3	1	2	2.25
Data leakage	DDPG	0	0	0	0	0
Host-based	DDPG	0	0	0	0	0
Intrusion anomalies	DDPG	3	3	1	3	2.5
Network-based	DDPG	0	0	0	0	0
Policy violation	DDPG	0	0	0	0	0
Synthetic traffic	DDPG	0	0	0	0	0
Unknown or Adaptive threats	DDPG	3	3	2	3	2.75
Adversarial threats	DDQN	0	0	0	0	0
AI-generated threats	DDQN	0	0	0	0	0
Behavioral	DDQN	3	1	1	2	1.75
Data leakage	DDQN	0	0	0	0	0
Host-based	DDQN	0	0	0	0	0
Intrusion anomalies	DDQN	0	0	0	0	0
Network-based	DDQN	0	0	0	0	0
Policy violation	DDQN	0	0	0	0	0
Synthetic traffic	DDQN	0	0	0	0	0
Unknown or Adaptive threats	DDQN	0	0	0	0	0
Adversarial threats	Deep SARSA	0	0	0	0	0
AI-generated threats	Deep SARSA	0	0	0	0	0
Behavioral	Deep SARSA	3	1	1	2	1.75
Data leakage	Deep SARSA	0	0	0	0	0
Host-based	Deep SARSA	0	0	0	0	0
Intrusion anomalies	Deep SARSA	0	0	0	0	0
Network-based	Deep SARSA	0	0	0	0	0
Policy violation	Deep SARSA	0	0	0	0	0
Synthetic traffic	Deep SARSA	0	0	0	0	0
Unknown or Adaptive threats	Deep SARSA	0	0	0	0	0
Adversarial threats	DQN	3	3	1	2	2.25
AI-generated threats	DQN	0	0	0	0	0
Behavioral	DQN	3	3	1	2	2.25
Data leakage	DQN	0	0	0	0	0
Host-based	DQN	0	0	0	0	0
Intrusion anomalies	DQN	3	3	3	3	3
Network-based	DQN	3	3	1	3	2.5
Policy violation	DQN	0	0	0	0	0
Synthetic traffic	DQN	0	0	0	0	0
Unknown or Adaptive threats	DQN	3	3	2	3	2.75
Adversarial threats	Dueling-DQN	0	0	0	0	0
AI-generated threats	Dueling-DQN	0	0	0	0	0
Behavioral	Dueling-DQN	3	1	1	2	1.75
Data leakage	Dueling-DQN	0	0	0	0	0
Host-based	Dueling-DQN	0	0	0	0	0
Intrusion anomalies	Dueling-DQN	0	0	0	0	0
Network-based	Dueling-DQN	0	0	0	0	0
Policy violation	Dueling-DQN	0	0	0	0	0
Synthetic traffic	Dueling-DQN	0	0	0	0	0
Unknown or Adaptive threats	Dueling-DQN	0	0	0	0	0
Adversarial threats	PPO	0	0	0	0	0

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Table A.6: SS for RL algorithms.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
AI-generated threats	PPO	0	0	0	0	0
Behavioral	PPO	0	0	0	0	0
Data leakage	PPO	0	0	0	0	0
Host-based	PPO	0	0	0	0	0
Intrusion anomalies	PPO	3	3	1	3	2.5
Network-based	PPO	3	3	3	2	2.75
Policy violation	PPO	0	0	0	0	0
Synthetic traffic	PPO	0	0	0	0	0
Unknown or Adaptive threats	PPO	3	3	3	3	3
Adversarial threats	Q-Learning	0	0	0	0	0
AI-generated threats	Q-Learning	0	0	0	0	0
Behavioral	Q-Learning	0	0	0	0	0
Data leakage	Q-Learning	0	0	0	0	0
Host-based	Q-Learning	0	0	0	0	0
Intrusion anomalies	Q-Learning	0	0	0	0	0
Network-based	Q-Learning	0	0	0	0	0
Policy violation	Q-Learning	0	0	0	0	0
Synthetic traffic	Q-Learning	0	0	0	0	0
Unknown or Adaptive threats	Q-Learning	3	1	1	2	1.75
Adversarial threats	RRL	0	0	0	0	0
AI-generated threats	RRL	0	0	0	0	0
Behavioral	RRL	0	0	0	0	0
Data leakage	RRL	0	0	0	0	0
Host-based	RRL	0	0	0	0	0
Intrusion anomalies	RRL	0	0	0	0	0
Network-based	RRL	3	1	1	3	2
Policy violation	RRL	0	0	0	0	0
Synthetic traffic	RRL	0	0	0	0	0
Unknown or Adaptive threats	RRL	0	0	0	0	0
Adversarial threats	SAC	0	0	0	0	0
AI-generated threats	SAC	0	0	0	0	0
Behavioral	SAC	0	0	0	0	0
Data leakage	SAC	0	0	0	0	0
Host-based	SAC	3	1	1	3	2
Intrusion anomalies	SAC	0	0	0	0	0
Network-based	SAC	0	0	0	0	0
Policy violation	SAC	0	0	0	0	0
Synthetic traffic	SAC	0	0	0	0	0
Unknown or Adaptive threats	SAC	0	0	0	0	0
Adversarial threats	SARSA	0	0	0	0	0
AI-generated threats	SARSA	0	0	0	0	0
Behavioral	SARSA	0	0	0	0	0
Data leakage	SARSA	0	0	0	0	0
Host-based	SARSA	0	0	0	0	0
Intrusion anomalies	SARSA	0	0	0	0	0
Network-based	SARSA	0	0	0	0	0
Policy violation	SARSA	0	0	0	0	0
Synthetic traffic	SARSA	0	0	0	0	0
Unknown or Adaptive threats	SARSA	3	1	1	2	1.75

End of table

Table A.7: SS for Gen AI algorithms.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
Adversarial threats	DALL·E	0	0	0	0	0
AI-generated threats	DALL·E	3	2	2	0	1.75
Behavioral	DALL·E	0	0	0	0	0
Data leakage	DALL·E	0	0	0	0	0
Host-based	DALL·E	0	0	0	0	0
Intrusion anomalies	DALL·E	3	2	1	0	1.5
Network-based	DALL·E	0	0	0	0	0
Policy violation	DALL·E	0	0	0	0	0
Synthetic traffic	DALL·E	0	0	0	0	0
Unknown or Adaptive threats	DALL·E	0	0	0	0	0
Adversarial threats	GAN	3	3	1	2	2.25

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Table A.7: SS for Gen AI algorithms.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
AI-generated threats	GAN	3	3	1	2	2.25
Behavioral	GAN	3	3	1	2	2.25
Data leakage	GAN	0	0	0	0	0
Host-based	GAN	0	0	0	0	0
Intrusion anomalies	GAN	3	3	1	1	2
Network-based	GAN	0	0	0	0	0
Policy violation	GAN	0	0	0	0	0
Synthetic traffic	GAN	0	0	0	0	0
Unknown or Adaptive threats	GAN	0	0	0	0	0
Adversarial threats	GAN + ANN	0	0	0	0	0
AI-generated threats	GAN + ANN	0	0	0	0	0
Behavioral	GAN + ANN	3	1	1	3	2
Data leakage	GAN + ANN	0	0	0	0	0
Host-based	GAN + ANN	0	0	0	0	0
Intrusion anomalies	GAN + ANN	0	0	0	0	0
Network-based	GAN + ANN	0	0	0	0	0
Policy violation	GAN + ANN	0	0	0	0	0
Synthetic traffic	GAN + ANN	0	0	0	0	0
Unknown or Adaptive threats	GAN + ANN	0	0	0	0	0
Adversarial threats	GAN + CNN	3	1	2	3	2.25
AI-generated threats	GAN + CNN	0	0	0	0	0
Behavioral	GAN + CNN	0	0	0	0	0
Data leakage	GAN + CNN	0	0	0	0	0
Host-based	GAN + CNN	0	0	0	0	0
Intrusion anomalies	GAN + CNN	0	0	0	0	0
Network-based	GAN + CNN	0	0	0	0	0
Policy violation	GAN + CNN	0	0	0	0	0
Synthetic traffic	GAN + CNN	0	0	0	0	0
Unknown or Adaptive threats	GAN + CNN	0	0	0	0	0
Adversarial threats	GAN + GPT	3	1	1	1	1.5
AI-generated threats	GAN + GPT	0	0	0	0	0
Behavioral	GAN + GPT	0	0	0	0	0
Data leakage	GAN + GPT	0	0	0	0	0
Host-based	GAN + GPT	0	0	0	0	0
Intrusion anomalies	GAN + GPT	0	0	0	0	0
Network-based	GAN + GPT	0	0	0	0	0
Policy violation	GAN + GPT	0	0	0	0	0
Synthetic traffic	GAN + GPT	0	0	0	0	0
Unknown or Adaptive threats	GAN + GPT	0	0	0	0	0
Adversarial threats	GAN + LSTM	3	3	2	2	2.5
AI-generated threats	GAN + LSTM	0	0	0	0	0
Behavioral	GAN + LSTM	3	3	1	2	2.25
Data leakage	GAN + LSTM	0	0	0	0	0
Host-based	GAN + LSTM	0	0	0	0	0
Intrusion anomalies	GAN + LSTM	3	3	1	2	2.25
Network-based	GAN + LSTM	3	3	1	3	2.5
Policy violation	GAN + LSTM	0	0	0	0	0
Synthetic traffic	GAN + LSTM	0	0	0	0	0
Unknown or Adaptive threats	GAN + LSTM	0	0	0	0	0
Adversarial threats	GAN + QLearning	0	0	0	0	0
AI-generated threats	GAN + QLearning	0	0	0	0	0
Behavioral	GAN + QLearning	0	0	0	0	0
Data leakage	GAN + QLearning	0	0	0	0	0
Host-based	GAN + QLearning	0	0	0	0	0
Intrusion anomalies	GAN + QLearning	3	1	1	3	2
Network-based	GAN + QLearning	0	0	0	0	0
Policy violation	GAN + QLearning	0	0	0	0	0
Synthetic traffic	GAN + QLearning	0	0	0	0	0
Unknown or Adaptive threats	GAN + QLearning	0	0	0	0	0
Adversarial threats	GAN + RF	0	0	0	0	0
AI-generated threats	GAN + RF	0	0	0	0	0
Behavioral	GAN + RF	3	2	1	3	2.25
Data leakage	GAN + RF	0	0	0	0	0
Host-based	GAN + RF	0	0	0	0	0
Intrusion anomalies	GAN + RF	0	0	0	0	0
Network-based	GAN + RF	3	2	2	3	2.5
Policy violation	GAN + RF	0	0	0	0	0
Synthetic traffic	GAN + RF	0	0	0	0	0
Unknown or Adaptive threats	GAN + RF	0	0	0	0	0
Adversarial threats	GAN + SVM	0	0	0	0	0
AI-generated threats	GAN + SVM	0	0	0	0	0

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Table A.7: SS for Gen AI algorithms.

Anomaly Category	Dataset	Relevance (R)	Coverage (C)	Usage (U)	Performance (P)	Support Score (SS)
Behavioral	GAN + SVM	3	2	1	3	2.25
Data leakage	GAN + SVM	0	0	0	0	0
Host-based	GAN + SVM	0	0	0	0	0
Intrusion anomalies	GAN + SVM	0	0	0	0	0
Network-based	GAN + SVM	3	3	1	2	2.25
Policy violation	GAN + SVM	0	0	0	0	0
Synthetic traffic	GAN + SVM	0	0	0	0	0
Unknown or Adaptive threats	GAN + SVM	0	0	0	0	0
Adversarial threats	GAN + Transformer	3	1	1	2	1.75
AI-generated threats	GAN + Transformer	0	0	0	0	0
Behavioral	GAN + Transformer	0	0	0	0	0
Data leakage	GAN + Transformer	0	0	0	0	0
Host-based	GAN + Transformer	0	0	0	0	0
Intrusion anomalies	GAN + Transformer	0	0	0	0	0
Network-based	GAN + Transformer	0	0	0	0	0
Policy violation	GAN + Transformer	0	0	0	0	0
Synthetic traffic	GAN + Transformer	0	0	0	0	0
Unknown or Adaptive threats	GAN + Transformer	0	0	0	0	0
Adversarial threats	Transformer	3	3	1	2	2.25
AI-generated threats	Transformer	0	0	0	0	0
Behavioral	Transformer	3	3	1	2	2.25
Data leakage	Transformer	0	0	0	0	0
Host-based	Transformer	0	0	0	0	0
Intrusion anomalies	Transformer	3	3	1	1	2
Network-based	Transformer	0	0	0	0	0
Policy violation	Transformer	0	0	0	0	0
Synthetic traffic	Transformer	0	0	0	0	0
Unknown or Adaptive threats	Transformer	0	0	0	0	0
Adversarial threats	VAE	3	3	1	2	2.25
AI-generated threats	VAE	0	0	0	0	0
Behavioral	VAE	3	3	1	2	2.25
Data leakage	VAE	0	0	0	0	0
Host-based	VAE	0	0	0	0	0
Intrusion anomalies	VAE	3	3	1	1	2
Network-based	VAE	0	0	0	0	0
Policy violation	VAE	0	0	0	0	0
Synthetic traffic	VAE	0	0	0	0	0
Unknown or Adaptive threats	VAE	0	0	0	0	0

End of table

Appendix B. Tables Relevant to RQ-2

Table B.1: Value-based RL algorithms used in cybersecurity.

Algorithm	Type	Action space	Policy	Family	Strengths	Limitations
Q-Learning (tabular)	Model-free	Discrete	Off-policy	Value-based	Very simple and interpretable baseline; useful for small state spaces and as a red-team baseline	Does not scale to high-dimensional states; weak generalization vs deep methods
SARSA (tabular)	Model-free	Discrete	On-policy	Value-based	On-policy updates can be steadier under non-stationarity; simple to implement; common baseline	Sample-inefficient; weaker asymptotic performance; sensitive to exploration policy
Deep SARSA	Model-free	Discrete	On-policy	Value-based	Improved multi-class intrusion detection (NSL-KDD, UNSW-NB15)	Detects only known attacks; less sample-efficient
DQN	Model-free	Discrete	Off-policy	Value-based	Baseline for discrete actions; replay buffer improves efficiency	Overestimation bias; poor with continuous actions; unstable under distribution shift
DDQN	Model-free	Discrete	Off-policy	Value-based	Mitigates DQN overestimation; stable with target network and replay	Discrete-only; sensitive to hyperparameters; underestimation possible
Dueling-DQN	Model-free	Discrete	Off-policy	Value-based	Separates $V(s)$ and advantage to improve value estimates; stronger than DQN in several comparisons	Still discrete-only; benefits depend on tuning; inherits replay/target sensitivities

Table B.2: Policy-based (policy gradient) RL algorithms used in cybersecurity.

Algorithm	Type	Action space	Policy	Family	Strengths	Limitations
PPO	Model-free	Continuous / Discrete	On-policy	Policy-gradient (SPG[1])	Stable clipped updates; widely reproducible across tasks	On-policy sample cost; sensitive to clip ϵ and GAE λ
A2C	Model-free	Discrete	On-policy	Policy-gradient (SPG[1])	Simple baseline; advantage estimation stabilizes updates	Lower performance than value-based DQN variants in ICS NIDS
A3C	Model-free	Continuous / Discrete	On-policy	Policy-gradient (SPG[1])	Asynchronous parallel workers speed wall-clock learning; strong baseline	On-policy sample inefficiency; stale-gradient effects; tuning entropy/LR needed

[1] Stochastic Policy Gradient (SPG);

Table B.3: Hybrid actor–critic RL algorithms used in cybersecurity.

Algorithm	Type	Action space	Policy	Family	Strengths	Limitations
Actor–Critic (generic)	Model-free	Continuous	Varies (on/off)	Actor–Critic	Combines critic stability with expressive policy; handles large action spaces	Coupled training can diverge; tuning entropy/advantage is critical
DDPG	Model-free	Continuous	Off-policy	Actor–Critic (DPG[2])	Continuous actions for fine-grained defense; replay improves data efficiency	Exploration brittle; sensitive to reward scaling; less stable than TD3/SAC
RRIoT[3]	Model-free	Continuous	Off-policy	Actor–Critic (DPG[2])	Handles sequential IoT traffic; recurrent encoder improves partial observability	Dataset imbalance; recurrent critics unstable; high compute cost
MAPPO	Model-free	Discrete	On-policy	Actor–Critic (SPG[1])	Shared critic accelerated convergence; outperformed IPPO in maritime OT	Simulation not fully realistic; reward–cost valuation limited
SAC	Model-free	Continuous	Off-policy	Actor–Critic (Max-Entropy[4])	High sample efficiency and stability via entropy regularization; robust for continuous control/response	Temperature tuning sensitivity; heavier compute/replay; less used for discrete-only tasks

[1] Stochastic Policy Gradient (SPG);
[2] Deterministic Policy Gradient (DPG);
[3] Recurrent Deep Deterministic Policy Gradient (RDDPG, as used in RRIoT);
[4] Soft Actor–Critic's maximum-entropy actor–critic formulation.

Table B.4: Summary of the state-of-the-art studies on RL for cybersecurity.

Ref.	Year	Dataset	RL algorithm	Objective	Results	Strengths	Limitations
[38]	2023	ToN–IoT (GPS / Modbus / Weather)	DQN	Red teaming / attack planning	Accuracy = 87.8% (Weather split)	Fast convergence; strong on Weather	Lower GPS accuracy (79.7%); Modbus weaker
[11]	2024	Smart grid simulator (custom)	DDQN (CNN & GRU)	Smart grid anomaly detection	Accuracy = 95.3%	Accurate minority attack detection; real-time adaptability	Simulation only
[39]	2023	NSL–KDD	Deep SARSA	Network intrusion detection	F1 = 84.6%	Superior minority-class detection	Evaluated on older datasets
[40]	2023	Not specified	Actor–Critic	Adaptive threat detection / response	Conceptual	Supports adversarial learning; real-time policy updates	No dataset or empirical validation
[41]	2023	IPMSRL (maritime OT simulation)	MAPPO & IPPO	Industrial / OT defense	Success rate = 99.5% (MAPPO)	Collaborative defense; configurable environment	Domain-specific; complex simulator setup
[42]	2023	IoT–23	Recurrent RL (GRU & DQN)	IoT device threat detection	F1 = 80.6%	Captures sequential IoT dynamics	High complexity; training overhead
[47]	2023	Traffic management (simulated)	DRL (PPO / DDPG)	IoT-based traffic & cyber threat response	Reward +38% over baseline	Multi-agent coordination; real-time safety decisions	Environment-specific; no open dataset
[48]	2023	Enterprise network (simulated)	DQN	Real-time network threat detection	Accuracy = 93.1%	Adaptive learning; dynamic policy updates	Limited generalizability to public benchmarks
[47]	2023	Smart city traffic (simulated)	DRL (policy gradient)	Smart city cyber–physical threat detection	Accuracy = 91.4%	Coordinated traffic–cyber response	Complex simulation; replication difficult
[49]	2023	CyberBattleSim	DQL (reward shaping)	Red team attack simulation & defense	Steps-to-goal: 27% faster convergence	Reward-optimized adaptive planning	Limited generalization beyond emulation

Appendix C. Tables Relevant to RQ-3

Table C.1: GAN-based generative AI algorithms used in cybersecurity.

Algorithm	Type	Output modality	Training paradigm	Family	Strengths	Limitations
Vanilla GAN	Generative	Network traffic / binaries	Adversarial (G vs D)	GAN	Simple baseline; effective for IDS data augmentation; learns distributional features	Training instability; mode collapse; limited scalability to high-dimensional domains
cGAN / ACGAN	Conditional GANs	Intrusion patterns, phishing emails	Adversarial (conditional labels)	GAN	Conditional control improves targeted attack synthesis and class balancing	Sensitive to label noise; prone to overfitting; requires rich labelled data
DCGAN / WGAN	Convolutional / Wasserstein GAN	Flow features, malware images	Adversarial (improved loss)	GAN	More stable training; WGAN alleviates mode collapse; DCGAN strong for feature learning	Higher computational cost; limited temporal modelling
Adaptive GAN	Adaptive / evolving GAN	Real-time threats	Adversarial (online updates)	GAN	Adapts to changing threat vectors with minimal retraining	No validation on distributed / real networks; assumes feature stability

Table C.2: VAE-based generative AI algorithms used in cybersecurity.

Algorithm	Type	Output modality	Training paradigm	Family	Strengths	Limitations
VAE	Probabilistic generative model	Traffic logs, anomaly features	Variational (ELBO)	VAE	Stable training; learns compact latent representations; useful for anomaly detection	Generates less realistic samples vs GANs; blurry / low-fidelity outputs

Table C.3: Transformer and LLM-based generative AI algorithms used in cybersecurity.

Algorithm	Type	Output modality	Training paradigm	Family	Strengths	Limitations
GPT / ChatGPT	Auto-regressive LLM	Text (logs, phishing, prompts)	Auto-regressive (self-attention)	Transformer / LLM	Strong at language-based cyber tasks (log parsing, phishing, social engineering)	Hallucinations; vulnerable to prompt injection
BERT / other transformers	Encoder-only transformer	Log data, alerts	Self-supervised	Transformer	Good for classification and embedding-based anomaly detection	Limited generative power; requires fine-tuning on domain data

Table C.4: Hybrid / emerging generative AI used in cybersecurity.

Algorithm	Type	Output modality	Training paradigm	Family	Strengths	Limitations
GAN + VAE	Hybrid generative	Traffic anomalies	Combined (adversarial + variational)	Hybrid	Balances VAE stability with GAN realism; better diversity + quality	Complex training; limited benchmarks; few cyber applications yet
GAN + Transformer	Hybrid adversarial + attention	Traffic / malware / logs	Combined (adversarial + attention)	Hybrid	Captures sequential + contextual features; promising for adaptive threats	Under-explored; reproducibility issues; training overhead
Diffusion models	Probabilistic denoising generative	Synthetic attack traffic	Denoising score matching	Diffusion	High-quality synthetic data; avoids mode collapse; robust for data augmentation	Very computationally expensive; little empirical validation in cybersecurity

Table C.5: Summary of the state-of-the-art studies on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations
[9]	GANs	Intrusion detection systems	CICIDS–2017; NSL–KDD; DARPA; CTU–13; RockYou	Taxonomy of GAN–based IDS approaches; highlights cGAN, DCGAN, WGAN for realistic intrusion patterns	Relies on legacy datasets (e.g., NSL–KDD); no real–world deployment or adversarial testing
[66]	GANs	Synthetic threat data generation	Simulated datasets	Uses GANs to generate synthetic threats for IDS; supports adversarial scenario emulation for training	Not validated on high–variance attack sets; no operational deployment
[72]	GANs	Network traffic anomaly detection	Not specified	Applies GAN–based models combining feature extraction and generation	No comparison to state–of–the–art DL; lacks adversarial defense integration
[21]	GANs	Survey on GANs for IDS	CICIDS–2017; NSL–KDD	Comprehensive survey; categorizes GAN types by threat model and dataset	Uses some outdated datasets; limited coverage of newer GAN variants or multimodal threats
[73]	GANs	IDS evaluation	NSL–KDD; CICIDS–2017	Evaluates WGAN, DCGAN, ACGAN with precision/recall	Dataset limitations; no longitudinal testing; GAN stability issues underexplored
[69]	GANs	IDS enhancement	CICIDS–2017	Proposes conditional GAN for attack pattern classification	No experimental implementation; minimal GAN configuration details
[74]	GANs	Phishing email defense	None	Generates phishing emails to harden classifiers via synthetic data	Email set lacks diversity; no comparison to transformer baselines
[75]	Adaptive GAN	Real–time threat detection	None	Adaptive GAN for evolving threats with minimal retraining	Not validated in dynamic/distributed environments; assumes feature stability
[76]	GANs, VAEs	Threat simulation (malware, phishing, intrusions)	Synthetic datasets	PCA–GAN hybrid for anomaly synthesis to train robust IDS	No real–time validation; limited dataset scope; no adversarial robustness checks
[67]	GANs, transformers	Anomaly detection	CICIDS–2017	Compares GANs vs transformers on intrusion data (detection rate, convergence, scalability)	Missing transformer tuning and GAN loss curves; reproducibility unclear
[68]	GANs, ChatGPT	Cyber threat analysis	None	Framework outlining Gen AI misuse scenarios and attack simulation potential	No implementation or experiments
[77]	GANs, GPT	GAN vs GPT in cybersecurity	CICIDS–2017; synthetic	Contrasts GAN and GPT for anomaly detection; reports accuracy/F1	Narrow dataset scope; simple GPT models; limited external validity
[78]	LLMs	Prompt injection, jailbreaking	None	Details jailbreak techniques and risks of unrestricted LLM use	No quantitative risk model; lacks detection/mitigation methods
[71]	LLMs	Defense simulation	None	Simulation architecture for cyber training using LLM outputs	Not evaluated; no red/blue team testing
[70]	LLMs, VAEs	Adaptive threat generation	None	LLM–driven adaptive response; feedback–based defense orchestration	No SIEM/SOAR integration; latency/performance metrics absent
[79]	ChatGPT	Explainable AI in cybersecurity	None	XAI–enhanced IDS with ChatGPT explanations; user–facing interpretability	Explanations not compared to ground truth/human baseline
[80]	ChatGPT	Social / ethical risks	None	Taxonomy of Gen AI misuse and threat classes; policy/ethics overview	Lacks empirical risk evaluation; conceptual focus

Continued on next page

Table C.5: Summary of the state-of-the-art studies on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations
[81]	ChatGPT	Explainable IDS	CICIDS–2017	Uses ChatGPT to explain IDS decisions with LLM–based rationales	No adversarial evaluation; runtime overhead not reported; reliance on closed APIs
[82]	ChatGPT, LLMs	Policy and regulation risk	None	Analysis of Gen AI risks; layered defense and audit–based mitigation framework	No implementation/validation
[10]	ChatGPT, LLMs	Automated hacking / defense	None	Conceptual analysis: AI–generated phishing, malware, social engineering	No threat modelling or mitigation strategies
[83]	ChatGPT, Bard	LLM capabilities for offence/defense	None	Compares ChatGPT vs Bard for attack vector generation (evasiveness, diversity)	Dataset unspecified; metrics mostly qualitative

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